

Python

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import pandas as pd
import matplotlib.pyplot as plt

def load_and_clean_data(file_path):
    """Loads CDC outbreak data and performs initial cleaning."""
    # Load dataset with expected columns: ['Year', 'State',
    'Disease', 'Cases', 'Population']
    df = pd.read_csv(file_path)

    # Remove rows with missing case counts to maintain data
    integrity
    clean_df = df.dropna(subset=['Cases'])

    # Standardize data by calculating incidence rate per 100,000
    residents
    clean_df['Incidence_Rate'] = (clean_df['Cases'] /
    clean_df['Population']) * 100000
    return clean_df

def analyze_disease_trend(df, disease_name):
    """Filters for a specific disease and calculates the annual
    mean incidence rate."""
    disease_filter = df[df['Disease'] == disease_name]
    return disease_filter.groupby('Year')['Incidence_Rate'].mean()

def visualize_trend(yearly_data, disease_name, output_file):
    """Generates and saves a professional line chart of the data
    trend."""
    plt.figure(figsize=(10, 6))
    plt.plot(yearly_data.index, yearly_data.values, marker='o',
    color='b', linewidth=2)

    # Chart styling and labeling
    plt.title(f'National {disease_name} Incidence Rate Over Time
    (Per 100k)', fontsize=14)
    plt.xlabel('Year', fontsize=12)
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plt.ylabel('Average Incidence Rate', fontsize=12)
plt.grid(True)

plt.savefig(output_file)
print(f"Analysis complete. Visualization saved to
{output_file}")

if __name__ == "__main__":
    # Configuration
    DATA_FILE = 'cdc_outbreak_data.csv'
    TARGET_DISEASE = 'Influenza'
    OUTPUT_IMAGE = 'flu_trend_analysis.png'

    # Execution pipeline
    processed_data = load_and_clean_data(DATA_FILE)
    annual_trend = analyze_disease_trend(processed_data,
TARGET_DISEASE)
    visualize_trend(annual_trend, TARGET_DISEASE, OUTPUT_IMAGE)
```